WORKED SOLUTION

STEP III 2013, Question 1

Pure • Half-Angle Substitution

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/13-s3-q1

The key idea

The half-angle substitution $t = \tan\frac{1}{2}x$ turns any rational function of $\sin x$ into a rational function of t, so the first integral becomes $\int \mathrm{d}t/(\text{quadratic})$, which we finish by completing the square. Getting the answer into the required form then needs the $\tan(A-B)$ formula. For I_n, the hint is designed so that the awkward denominator cancels: $\sin^{n+1}x + 2\sin^n x = \sin^n x\,(2+\sin x)$. That gives a recurrence which climbs from I_0, the only integral that genuinely needs the first part.

Solution

The substitution

With $t = \tan\frac{1}{2}x$, the chain rule gives

$$\frac{\mathrm{d}t}{\mathrm{d}x} = \tfrac{1}{2}\sec^2\tfrac{1}{2}x = \tfrac{1}{2}(1+\tan^2\tfrac{1}{2}x) = \tfrac{1}{2}(1+t^2).$$

For the sine, use the double-angle formula and then write $\cos^2\frac{1}{2}x = 1/\sec^2\frac{1}{2}x$:

$$\sin x = 2\sin\tfrac{1}{2}x\cos\tfrac{1}{2}x = 2\tan\tfrac{1}{2}x\cos^2\tfrac{1}{2}x = \frac{2\tan\frac{1}{2}x}{\sec^2\frac{1}{2}x} = \frac{2t}{1+t^2}.$$

Both results are shown.

The first integral

As x runs from 0 to $\frac{1}{2}\pi$, t runs from $\tan 0 = 0$ to $\tan\frac{1}{4}\pi = 1$, and $\mathrm{d}x = \dfrac{2\,\mathrm{d}t}{1+t^2}$. Also

$$1 + a\sin x = \frac{1+t^2+2at}{1+t^2},$$

so the factors $1+t^2$ cancel and

$$\int_0^{\frac{1}{2}\pi} \frac{\mathrm{d}x}{1+a\sin x} = \int_0^1 \frac{2\,\mathrm{d}t}{t^2+2at+1} = \int_0^1 \frac{2\,\mathrm{d}t}{(t+a)^2+(1-a^2)}.$$

Since $0<a<1$, the constant $1-a^2$ is positive; write $r=\sqrt{1-a^2}$. Using $\int \mathrm{d}u/(u^2+r^2) = \frac{1}{r}\arctan\frac{u}{r}$,

$$\int_0^{\frac{1}{2}\pi} \frac{\mathrm{d}x}{1+a\sin x} = \frac{2}{r}\Big[\arctan\frac{t+a}{r}\Big]_0^1 = \frac{2}{r}(\alpha-\beta),$$

where

$$\tan\alpha = \frac{1+a}{r} = \frac{1+a}{\sqrt{1-a}\sqrt{1+a}} = \frac{\sqrt{1+a}}{\sqrt{1-a}}, \qquad \tan\beta = \frac{a}{r},$$

with $\alpha,\beta\in(0,\frac{1}{2}\pi)$.

Ask yourself This doesn't look like the answer we want. How do we turn a difference of two arctangents into a single one?

Take the tangent of the difference. We have

$$\tan\alpha-\tan\beta = \frac{(1+a)-a}{r} = \frac{1}{r}, \qquad 1+\tan\alpha\tan\beta = 1+\frac{a(1+a)}{1-a^2} = 1+\frac{a}{1-a} = \frac{1}{1-a},$$

so

$$\tan(\alpha-\beta) = \frac{\tan\alpha-\tan\beta}{1+\tan\alpha\tan\beta} = \frac{1-a}{r} = \frac{1-a}{\sqrt{1-a}\sqrt{1+a}} = \frac{\sqrt{1-a}}{\sqrt{1+a}}.$$

We need $\alpha-\beta$ to lie in $(-\frac{1}{2}\pi,\frac{1}{2}\pi)$, the range of arctan, before we can invert. It does: both angles are in $(0,\frac{1}{2}\pi)$, and $\tan\alpha>\tan\beta$ (their difference is $1/r>0$), so $0<\alpha-\beta<\frac{1}{2}\pi$. Hence $\alpha-\beta=\arctan\dfrac{\sqrt{1-a}}{\sqrt{1+a}}$ and

$$\int_0^{\frac{1}{2}\pi} \frac{\mathrm{d}x}{1+a\sin x} = \frac{2}{\sqrt{1-a^2}}\arctan\frac{\sqrt{1-a}}{\sqrt{1+a}},$$

as required.

Evaluating I_3

Following the hint,

$$I_{n+1}+2I_n = \int_0^{\frac{1}{2}\pi} \frac{\sin^n x\,(\sin x+2)}{2+\sin x}\,\mathrm{d}x = \int_0^{\frac{1}{2}\pi} \sin^n x\,\mathrm{d}x.$$

We need the right-hand side for $n=0,1,2$:

$$\int_0^{\frac{1}{2}\pi} 1\,\mathrm{d}x = \tfrac{1}{2}\pi, \qquad \int_0^{\frac{1}{2}\pi} \sin x\,\mathrm{d}x = 1, \qquad \int_0^{\frac{1}{2}\pi} \sin^2 x\,\mathrm{d}x = \int_0^{\frac{1}{2}\pi} \tfrac{1}{2}(1-\cos 2x)\,\mathrm{d}x = \tfrac{1}{4}\pi.$$

Ask yourself The recurrence links each I_n to the next, but where does it start?

We need I_0 separately, and this is where the first part comes in. Taking out a factor of 2 and using $a = \frac{1}{2}$,

$$I_0 = \int_0^{\frac{1}{2}\pi} \frac{\mathrm{d}x}{2+\sin x} = \frac{1}{2}\int_0^{\frac{1}{2}\pi} \frac{\mathrm{d}x}{1+\frac{1}{2}\sin x} = \frac{1}{2}\cdot\frac{2}{\sqrt{3/4}}\arctan\frac{\sqrt{1/2}}{\sqrt{3/2}}$$
$$= \frac{2}{\sqrt{3}}\arctan\frac{1}{\sqrt{3}} = \frac{2}{\sqrt{3}}\cdot\frac{\pi}{6} = \frac{\pi}{3\sqrt{3}}.$$

Now climb the recurrence $I_{n+1} = \int_0^{\frac{1}{2}\pi} \sin^n x\,\mathrm{d}x - 2I_n$:

$$I_1 = \tfrac{1}{2}\pi - 2I_0 = \frac{\pi}{2} - \frac{2\pi}{3\sqrt{3}},$$
$$I_2 = 1 - 2I_1 = 1 - \pi + \frac{4\pi}{3\sqrt{3}},$$
$$I_3 = \tfrac{1}{4}\pi - 2I_2 = \frac{\pi}{4} - 2 + 2\pi - \frac{8\pi}{3\sqrt{3}}.$$

Therefore

$$I_3 = \frac{9\pi}{4} - \frac{8\sqrt{3}\,\pi}{9} - 2.$$

Checks

- As $a \to 0$ the first result should give $\int_0^{\frac{1}{2}\pi} 1\,\mathrm{d}x = \frac{1}{2}\pi$; indeed it tends to $2\arctan 1 = \frac{1}{2}\pi$.
- Since $0 \le \sin^3 x/(2+\sin x) \le \frac{1}{3}$ on $[0, \frac{1}{2}\pi]$, we need $0 < I_3 < \frac{1}{6}\pi \approx 0.52$. Our answer is $I_3 \approx 0.2318$, and numerical integration of I_3 agrees to many decimal places. Similarly $I_0 = \pi/(3\sqrt{3}) \approx 0.6046$, which lies between $\frac{\pi}{6}$ and $\frac{\pi}{4}$ as it must.
- The values decrease, $I_0 > I_1 > I_2 > I_3 > 0$, as they should, because $0 \le \sin x \le 1$ on the interval.

Common slips

- Forgetting to change the limits to $t = 0$ and $t = 1$, or forgetting the factor $2/(1+t^2)$ from $\mathrm{d}x$.
- Using $\tan(\alpha - \beta)$ to conclude $\alpha - \beta = \arctan(\dots)$ without checking that $\alpha - \beta$ lies in $(-\frac{1}{2}\pi, \frac{1}{2}\pi)$.
- Applying the first result to I_0 with $a = 2$: it only holds for $0 < a < 1$, so the factor 2 must be taken out first.

Beyond STEP

The $t = \tan\frac{1}{2}x$ substitution (often called the Weierstrass substitution) shows that every rational function of $\sin x$ and $\cos x$ has an elementary antiderivative, since partial fractions handle any rational function of t. Geometrically it is the stereographic parametrisation of the unit circle, $(\cos x, \sin x) = \left(\frac{1-t^2}{1+t^2}, \frac{2t}{1+t^2}\right)$, which also produces every Pythagorean triple from rational t. Integrals like $\int_0^{2\pi} \mathrm{d}x/(1 + a\sin x)$ are also a classic first application of contour integration in complex analysis; Needham's *Visual Complex Analysis* is a beautiful place to see why.

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